

Origins of elasticity in molecular materials

Received: 27 November 2022

Accepted: 3 January 2025

Published online: 21 February 2025



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Elasticity is ubiquitous and produces a spontaneously reversible response to applied stress¹. Despite the utility and importance of this property in regard to scientific and engineering applications, the atomic-scale location of the force that returns an object to its original shape remains elusive in molecular crystals. Here we use a series of density functional theory calculations to locate precisely where the energy is stored when single crystals of three molecular materials are placed under elastic stress. We show for each material that different intermolecular interactions are responsible for the restoring force under both expansive and compressive strain. These findings provide insight into the elastic behaviour of crystalline materials that is needed for more efficient design of flexible technologies and future smart devices.

Elasticity is both ubiquitous and fundamental to the cellular basis of life^{2,3} and technology^{4,5}. It underpins the way that animals move⁶ and, crucially, allows the design and construction of everything from optical fibres⁷ to skyscraper buildings⁸, because elasticity produces a spontaneously reversible response to applied stress without catastrophic failure¹. Despite the utility and importance of this property in regard to myriad engineering applications and millennia of use⁹, the molecular origin of the restoring force remains elusive¹⁰, limiting the development of advanced technologies that rely on elasticity. Here we locate precisely where the energy is stored when single crystals of bis(acetylacetonato)copper(II) ([Cu(acac)₂], [CuL¹]₂) are placed under elastic stress, and elucidate the anisotropic mechanism that allows the spontaneous return of these crystals to their original shape. We demonstrate that π – π interactions are responsible for the restoring force under expansive strain, while dispersive interactions produce the restoring force under compressive strain. We then show that different combinations of interactions are responsible for the restoring force in two additional compounds, [CuL²]₂ and [CuL³]₂, which differ from [CuL¹]₂ by the substitution of methyl groups with ethyl and propyl groups, respectively. Atomic-scale insight into the elastic behaviour of such materials will allow more efficient design of new building materials and future smart devices.

All materials display some amount of elastic response to applied stress. When an elastically flexible object is bent by the application

of force, parts of that object are compressed and other parts expand. When the stress causing an elastic deformation is released, the object spontaneously returns to its original shape^{1,11}. This allows deformation without fracture—a property that makes elastic materials exceptionally useful. For the object to return to its original shape, energy is required that is stored in the form of strain within the material; the release of this strain produces the required force to recover the shape of the object. While it is generally understood that the origin of this restoring force is interatomic and, in molecular crystals intermolecular^{1,12–16}, the restoring force is typically regarded as a bulk material property with multiple interactions combining to yield the observed behaviour¹. In molecular crystals there is negligible deformation of covalent bonds in the elastic regime^{17,18} and thus elastic behaviour is governed by weak interactions. The atomic-scale mechanisms of elastic deformation in molecular crystals therefore are likely to be different from those in non-molecular materials. The recent determination of the molecular crystal structures of elastically deformed single crystals at high precision provides a unique opportunity to locate the atomic origin of the restoring force within an elastic material. The identification and quantification of these restoring forces could lead the way for the development of new flexible technologies such as optical waveguides¹⁹, mechanomagnets²⁰ and components in photonic circuits^{21–24}.

[CuL¹]₂ is known to have appreciable elastic flexibility in the crystalline state²⁵ (Fig. 1); these crystals can be repeatedly bent and unbent

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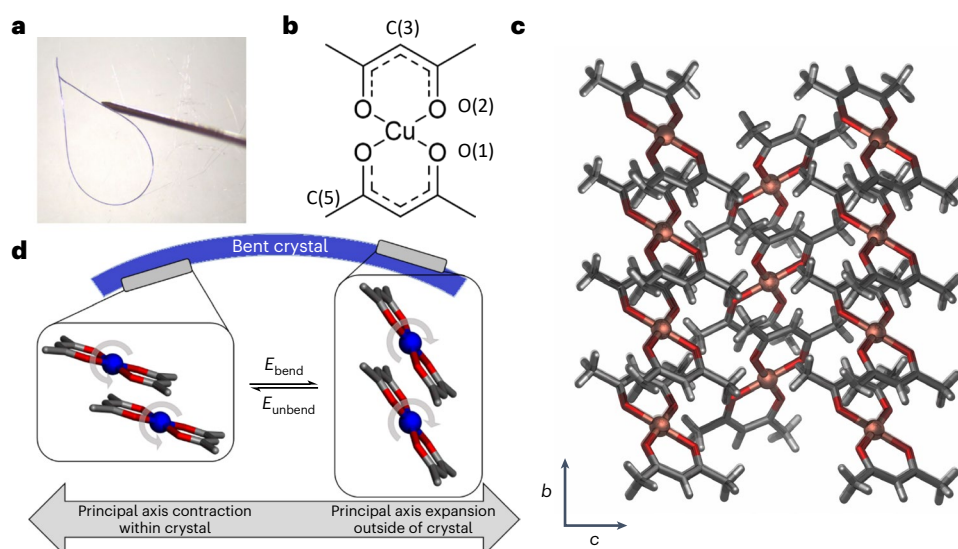


Fig. 1 | Crystal structure and elastic bending properties of $[\text{CuL}_2]$. **a**, An elastically bent crystal of $[\text{CuL}_2]$. **b**, Chemical structure of $[\text{CuL}_2]$, with numbering scheme. **c**, Crystal packing when viewed along the crystallographic a axis. **d**, The molecular-scale bending mechanism of the crystal; molecules

reversibly rotate and slip to form the macroscopic deformation when bent. E_{bend} is the energy required to elastically bend a crystal and E_{unbend} is the energy released when a bent crystal is elastically straightened.

without loss of crystallinity. Microfocused diffraction experiments recently demonstrated that this notable flexibility is permitted by molecular rotations that cause expansion of the crystal on the outer edge of the curve, and compression on the inner edge (Fig. 1). Here we use the explicit deformation information provided by this mechanistic knowledge to understand the energetic changes that occur when a crystal is subjected to stress, allowing the precise determination of the location of the elastic restoring force and the mechanism that returns these crystals to their original shape.

The crystal structure of $[\text{CuL}_2]$ (monoclinic, $P2_1/n$) consists of chains of planar molecules that propagate in the $[010]$ direction^{20,25,26} (Fig. 1). Energy framework calculations^{27–29} (Fig. 2) were used to estimate the relative strength of intermolecular interactions in the structure. Three important interactions were identified and analysed using the structure of the unbent crystal: intrachain π -stacking (along $[010]$) and two approximately orthogonal interchain interactions between adjacent molecules related by the n -glide plane, which we term CH- π (between molecules along $[101]$) and CH-O (between molecules along $[10\bar{1}]$) (Fig. 2). The π -stacking is a combination of both offset face-to-face π - π and Cu- π interactions and has an energy of around -80 kJ mol^{-1} . CH- π is best described as an edge-face interaction between the γ -proton (H(3)) and γ -carbon (C(3); Fig. 1) of an adjacent molecule, and is substantially weaker (by about -13 kJ mol^{-1}) than the π -stacking interaction. CH-O is the weakest interaction (approximately -10 kJ mol^{-1}) and consists of methyl (C(5)) proton-oxygen interactions (O(1) and O(2); Fig. 1).

The established mechanism of elastic bending in $[\text{CuL}_2]$ (ref. 25) was then used to investigate the way in which the energies of these interactions change as a function of deformation. Because elastic deformation generally produces a linear relationship between stress and strain¹, 170 structures were generated using linear extrapolation (Supplementary Information) under various levels of deformation, including hypothetical structures beyond the measured elastic limit of the material ($\sim 1\%$ strain, indicated by purple bars in each panel of Fig. 3). Counterpoise corrected³⁰ density functional theory (DFT) with the ωB97xD (ref. 31) functional was used to calculate the energies of the three most influential interactions (π - π , CH- π and CH-O) at different degrees of deformation. To validate this method of modelling the energy of the crystal via pairs of dimers, the energy of the entire crystal

structure was also calculated using DFT³² with the B3LYP functional^{33,34} and the DFT-D3 method using the Becke-Johnson damping function³⁵. The resulting energies are summarized in Fig. 3. To demonstrate the general applicability of this method, similar calculations were also performed on two chemically related materials ($[\text{CuL}_2]$ and $[\text{CuL}_3]$, which differ from $[\text{CuL}_2]$ by the substitution of the methyl groups with ethyl and propyl groups, respectively), and that have been determined to have different mechanisms of bending³⁶ (Supplementary Figs. 8 and 10).

Interestingly, the minima of each individual interaction in $[\text{CuL}_2]$ were found at positions well outside the elastic limit of the crystal (~ 2.1 , 4.2 and 11.4% strain for π - π , CH- π and CH-O, respectively (Fig. 3d)). Furthermore, the elastic region of the material as a whole is much smaller than that of the individual interactions, and the sum of the interactions reproduces the observed equilibrium structure. This shows that, although weak compared with the π - π interaction, the CH- π and CH-O interactions play a crucial role in determining the structure and elastic interactions in $[\text{CuL}_2]$. As none of the interactions are minimized in the lowest total energy structure, the elastic interactions in $[\text{CuL}_2]$ are frustrated. Such interactions are important in quantum magnetism³⁷ and spin cross-over materials^{38,39}, where they are known to lead to a wide range of exotic phenomena and generically lead to ‘softness’ in the state of the system; the frustrated elastic interactions in $[\text{CuL}_2]$ may be related to its flexibility.

Each of the individual interactions can be fitted to a Morse potential (Fig. 3a–c), which correlates well with the energy framework calculations. The π - π interaction has a potential well depth (difference between dissociation energy and minimum) of 65.9 kJ mol^{-1} and a spring constant of 22.9 N m^{-1} . Spring constants for the CH- π and CH-O interactions are 25.9 and 24.0 N m^{-1} with potential well depths of 10.6 and 7.8 kJ mol^{-1} , respectively (Supplementary Tables 5 and 6). Thus, the CH- π interaction is harder and deeper than the CH-O potential with respect to distance. This is consistent with previous reports^{25,40} that the $[101]$ face of the crystal has a smaller elastic modulus and undergoes greater deformation across the bent crystal than the $(10\bar{1})$ direction (the direction of CH-O interactions). The well depths found for $[\text{CuL}_2]$ are similar in magnitude to those found for $[\text{CuL}_1]$, although the π - π potential is much narrower (well depth 88.9 kJ mol^{-1} , spring constant 1.75 N m^{-1}), while $[\text{CuL}_3]$ was found to be much stiffer (spring constants

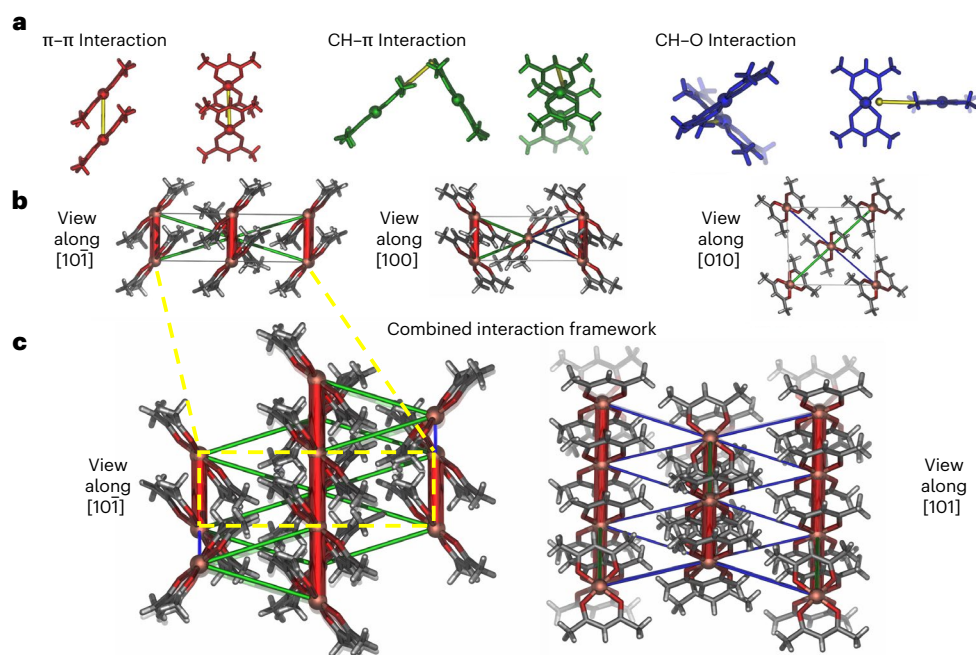


Fig. 2 | Energy framework analysis of $[\text{CuL}_2]$. **a**, The three key interactions identified in the crystal structure of $[\text{CuL}_2]$. **b**, Three views of the unit cell of $[\text{CuL}_2]$ highlighting the energy framework. **c**, The combined energy framework

of $[\text{CuL}_2]$. **b, c**, Red, green and blue tubes correspond to interactions π - π , CH- π and CH-O, respectively. Tube thickness represents the relative magnitude of interaction energies.

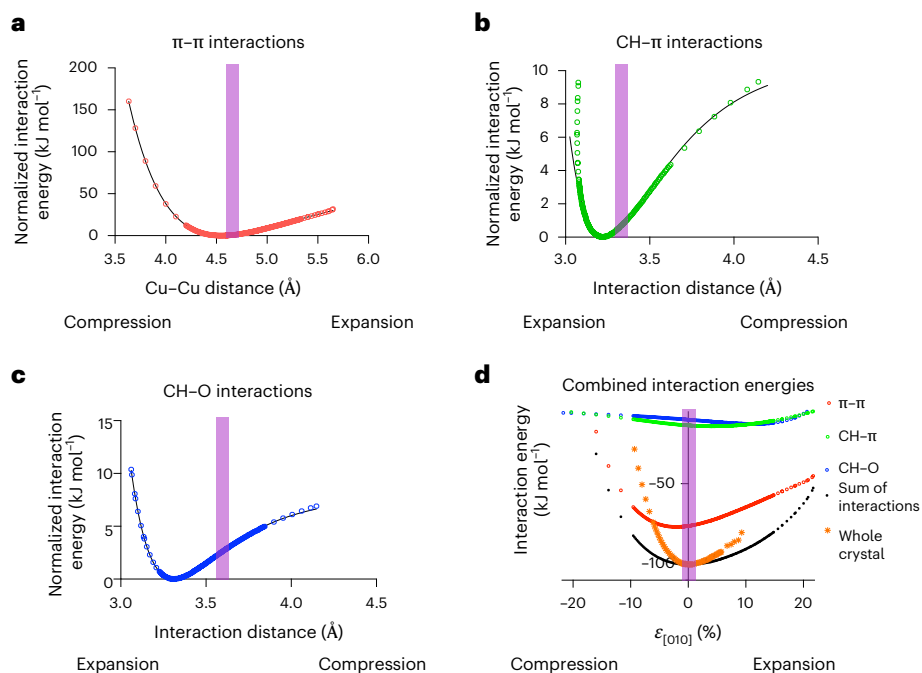


Fig. 3 | Change in energy in response to bending in $[\text{CuL}_2]$. **a–d**, Purple bars represent the range of experimental data used for extrapolation, which approximates the measured elastic limit of the crystal ($\pm 1\%$ strain). **a**, Variation in π - π interaction energy with distance. **b**, Variation in CH- π interaction energy with distance (defined by H(3)...C(3) separation). **c**, Variation in CH-O interaction energy with distance (defined by C(5)...centroid of O(1) and O(2) separation). **a–c**, In each case, interaction energies are plotted with the minimum of the well set to 0. Solid curves represent fits to a Morse potential (Supplementary Information). Dissociation of dimers at long interaction distances occurs on the outside of the bent crystal for π - π interactions, and on the inside for

CH- π and CH-O interactions (indicated by expansion and compression, respectively). **d**, Different interaction energies compared with deformation of the crystallographic b axis of $[\text{CuL}_2]$ (ϵ). Interaction energies of the three dimer interactions are summed and shown (black curve), as well as energy calculations for the entire crystal structure (Supplementary Information). The left-hand side of the graph represents compression (inside of the bent crystal) and the right-hand side expansion (outside of the bent crystal). Plots of CH- π and CH-O interaction energy with respect to copper–copper separations are given in Supplementary Figs. 3 and 4.

>600 N m⁻¹; Supplementary Table 6), consistent with previous observations that each of these materials has different molecular mechanisms of elastic deformation³⁶ and suggesting that the introduction of aliphatic chains into these molecular crystals tends to decrease their mechanical flexibility, because aliphatic groups can yield an increased number of sites for CH–HC interactions, which are generally repulsive⁴¹.

A subtle cancellation between the contributions of the largest intermolecular interactions in each structure leads to a minimum in the sum of these interaction energies that coincides with the undeformed crystal structure (Fig. 3d and Supplementary Figs. 8 and 10). That is, the sum of these intermolecular interactions predicts the equilibrium crystal structure. Calculations for the full crystal also place the minimum at the same position, implying that other pairwise interactions are not influential in the elastic region (Fig. 3d). Importantly, in the elastic region, the shape and curvature of the molecular and crystalline calculations agree well, demonstrating that the three interactions studied dominate the elastic energy for small strains. The deviation between the sum of molecular potentials and full crystalline calculation energies at larger strains suggests that longer-range interactions, or interactions between trimers, tetramers or larger oligomers, become important in this regime along with changes in unit cell dimensions. For instance, under high degrees of strain, even if the individual dimer's interactions become more stabilizing, deformation in the overall structure results in other long-range interactions becoming unfavourable. These oligomeric effects result in a steeper potential well compared with the individual dimers and become more influential the further the crystal is from neutral strain. These effects, however, are important only beyond the experimentally observed limit of elastic deformation, where other deformation mechanisms are expected to operate.

The above calculations provide detailed insight into the way that interactions change under both compressive and expansive strain, and allow precise determination of the origin of the restoring force in elastically deformed crystals. For example, when a crystal of [CuL₂] is bent to its elastic limit (1% deformation of the *b* axis), the [010] axis of the crystal expands on the outside of the crystal, increasing the energy of the pairwise π – π interaction (Fig. 3a) by 0.29 kJ mol⁻¹, while simultaneously lowering the CH– π and CH–O interactions by a combined 0.27 kJ mol⁻¹ (Fig. 3b,c). The change in π – π interaction energy is 0.02 kJ mol⁻¹ more than the simultaneous stabilization of the CH– π and CH–O interactions. Thus, on the outside of the crystal the potential energy is located in the π – π interactions. The opposite occurs on the inside of the crystal where the [010] axis is compressed, the energy of the π – π interaction decreases by 0.22 kJ mol⁻¹ (Fig. 3a) and CH– π and CH–O interactions increase by 0.29 kJ mol⁻¹. This results in 0.07 kJ mol⁻¹ of potential energy located in the CH– π and CH–O interactions. The restoring force on the outside of a bent loop of [CuL₂] therefore originates within the π – π interaction, while on the inside it originates in the combination of CH– π and CH–O interactions. The π – π interactions play a similar role on the outside of a bent loop in [CuL₂] and [CuL₃], while the other interactions in both of these materials are far less influential than the CH– π and CH–O interactions in [CuL₂].

In [CuL₂] the forces originating from π – π interaction largely cancel those from CH– π and CH–O interactions; because these forces have opposite sign and similar magnitude for both compression and expansion, the total elastic interaction (black curve in Fig. 3d) is much softer than any of the individual interactions that comprise the crystal: that is, the cancellation of π – π forces by CH– π and CH–O forces greatly enhances the flexibility of [CuL₂]. Nevertheless, on the inside of a crystal bent to the limit of elastic deformation, the elastic energy stored within the crystal is equal to approximately 70 J mol⁻¹ (for a 1% deformation of the *b* axis). On the outside of the same crystal, the stored elastic energy is equal to approximately 20 J mol⁻¹, making the total stored elastic potential energy within a crystal of [CuL₂] bent to its elastic limit approximately 90 J mol⁻¹; a similar amount of

energy is stored within [CuL₃]. While this is a small amount of energy compared with typical chemical processes, it is comparable to many mechanical energy scales. Each mole of [CuL₂] would be able to lift a 9 kg weight to a height of 1 m against gravity using the energy stored within the intermolecular interactions on bending; in other words, a crystal of [CuL₂] would be able to lift a mass approximately 30 times its own weight 1 m against gravity. To experimentally demonstrate these findings (in part), a series of single-crystal cantilevers were prepared in which steel balls of varying size (up to 100 times the weight of the crystal) were attached to single crystals of [CuL₂]. Application of force to the steel ball resulted in the crystal bending and, once released, the ball was raised higher against gravity than its equilibrium position (Supplementary Videos 1–4).

These results also provide further insight into the mechanism of plastic deformation of [CuL₂] observed beyond the elastic limit²⁵. Slip-page of planes⁴² lying parallel to (010) would result in CH– π interactions being required to break and reform, causing permanent deformation. This is facilitated by the relatively shallow potential well of this interaction compared with the π -stacking interaction. Crystals of [CuL₂] have also been observed to fibrillate along the [010] direction under applied force, which is consistent with the weak nature of CH– π and CH–O interactions.

Changes in nearest-neighbour interaction energies across elastically bent crystals reveal the origin of the elastic restoring force. In [CuL₂], the π -stacking interaction energy weakens on the expanded edge of the crystal while CH– π and CH–O interactions are strengthened. The opposite occurs on the contracted edge of the crystal. Those interactions that are destabilized result in an increase in the overall potential energy of the system. Therefore, because regions of the bent crystal undergoing expansion are restored via the π -stacking interaction, regions under compression are restored via CH– π and CH–O interactions. By locating the restoring force within elastically flexible materials, we provide molecular-level insights that cannot be gained from the elastic modulus alone. This method is applicable to any system in which the mechanism of elastic deformation is known with sufficient precision; following identification of key interactions, crystal structures with known trends of elastic deformation can be extrapolated to pinpoint the location of the restoring force. Such outcomes open the door for the design of new and elastically flexible crystals for advanced technologies.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41563-025-02133-w>.

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Methods

Crystal structures from the literature were modified to fix hydrogen atoms in geometrically optimal positions, and were refined using SHELXL⁴³. Interaction energies, lattice energies and energy frameworks were calculated using CrystalExplorer 17.5 (refs. 27,28) at the CE-B3LYP/6-31G(d,p) level. Structures were extrapolated using python scripts (Supplementary Information). Molecular calculations for [CuL₂¹] were performed using DFT at the unrestricted formalism, with the ωB97xD (ref. 31) functional employed for in-depth dimeric pairs intermolecular energies study of [CuL₂¹] in the solid state. In all calculations, a split-valence, triple-zeta basis set (6-311 + + G(d,p))^{44,45} was used for C, H and O atoms, and a double-zeta electrochemical potential basis set (LanL2DZ)⁴⁶ for copper atoms. We used the crystallographic and extrapolated coordinates as input coordinates, without relaxed optimization, for the theoretical analysis of non-covalent interactions present in the solid-state system. Calculations were performed using Gaussian16 v.B.01 (ref. 47). Interaction energies were calculated using the counterpoise method³⁰, which provides correction for basis set superposition error. An ultrafine grid was applied for all calculations. Graphical output from calculations was plotted with Igor (Wavemetrics, v.6.1.1.2) and GraphPad Prism⁴⁸. Calculations for the full crystal structure were performed using the Vienna ab initio simulation package (VASP) v.5.4.4 (ref. 49), with a set of crystal structures from the crystallographic measurements and extrapolated coordinates (Supplementary Information). We used the B3LYP functional^{33,34} with the DFT-D3 method and the Becke–Johnson damping function³⁵. The cutoff energy of plane-wave expansion was set to 500 eV and the convergence criterion for energy was 10^{−6} eV. The Brillouin zone was sampled at the Γ-point. Structures were optimized while holding both cell shape and volume fixed. Details of the additional calculations for [CuL₂²] and [CuL₂³] are similar and are presented in Supplementary Information.

Data availability

The results of calculations are provided as Supplementary Data 1–4. Source data are provided with this paper.

Code availability

The Python scripts used to produce the extrapolated atomic coordinates as described in Supplementary Information are available as Supplementary Code 1 or from the authors.

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Acknowledgements

A.J.T. thanks AINSE Limited for providing financial assistance (PGRA Award). B.S.K.C. thanks the University of Queensland for Research Training Program financial support. The University of Queensland Research Computing Centre, Phoenix HPC service at the University of Adelaide and Pawsey Supercomputing Research Centre through the National Computational Merit Allocation Scheme are thanked for providing high-performance, computing resources. This research was supported by the Australian Government's National Collaborative Research Infrastructure Strategy (NCRIS). B.J.P. thanks the Australian Research Council for financial support (DP181006201). J.D.E. is a recipient of an Australian Research Council Discovery Early Career Award (DE220100163). We thank A. Grosjean for providing input into the design of CrystalExplorer calculations, and for comments on an early version of the paper.

Author contributions

B.S.K.C. performed DFT calculations for [CuL₂¹]. A.J.T. wrote the Python scripts for structure extrapolation and undertook primary data analysis. E.P.K. performed VASP calculations. J.A.P. and A.J.T. performed CrystalExplorer calculations. J.D.E. performed calculations for [CuL₂²] and [CuL₂³]. B.J.P., M.A.S., J.C.M. and J.K.C. directed the research. All authors contributed to the analysis of results and writing of the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41563-025-02133-w>.

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Peer review information *Nature Materials* thanks Shotaro Hayashi, Satoshi Takamizawa and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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